LYDIA PEDERSEN

SOLUTIONS ENGINEER · SIGNAL INTEGRITY & SYSTEMS ARCHITECTURE

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PROFESSIONAL SUMMARY

Solutions Engineer with 20 years of experience across high-speed physical-layer validation, production AI pipeline architecture, and technical communication. I focus on the measurement methodologies and physical-layer standards that enable hardware infrastructure to support modern computational workloads. I have extensive experience in high-speed serial interface characterization and agentic AI automation of governed, repeatable workflows. I approach new specifications and architectures from first principles to ramp quickly on evolving product portfolios and help teams translate complex validation challenges into clear technical and business decisions.

PROFESSIONAL EXPERIENCE

Solutions Engineer · Keysight Technologies · Everett, WA — Nov 2026 – Present

Partner with field sales and engineering customers in the Greater Seattle region to configure high-speed digital and optical test solutions across oscilloscopes, BERTs, AWGs, protocol analyzers, VNAs, and optical measurement portfolios.

- Translate complex customer measurement and compliance requirements into integrated hardware and software workflows, and deliver technical demonstrations at customer sites and seminars.
- Act as a technical liaison between customers and internal R&D, marketing, and support teams, surfacing field requirements to inform product roadmaps.

Senior Technical Writer · Gravitee · Remote — Jun 2023 – Jul 2026

Wrote and maintained developer documentation for an enterprise API management platform, including Kafka and Server-Sent Events integration guides, the expression language reference, Docker deployment, and gateway resource sizing. Interviewed engineers to capture system design decisions, verified configurations hands-on, and built four production AI automation pipelines through agentic and prompt engineering patterns.

- **AI Doc Writer Agent:** Architected a multi-stage pipeline that converted PR diffs and Jira tickets into publication-ready documentation using fidelity scoring, Best-of-N evaluation with adaptive scaling, a hallucination guard cascade, and multi-stage recovery, validated with deduplication metrics and cross-file consistency checks.
- **Analytics Deflection Engine:** Refactored a monolithic script into a modular, product-agnostic pipeline that transformed raw AI assistant analytics into ranked, privacy-safe documentation improvement reports using parameter-sweep self-correction, multi-signal priority scoring (impact, pain, breadth), a k-anonymity privacy model, and automated Jira task generation.
- **Link Governance Engine:** Engineered a multi-category link classifier with safety-gated autofix, page-version index validation, and external link health checks that automated broken-link remediation across versioned documentation sites with auditable decisions.
- **Semantic Redirect Engine:** Built an Extract, Embed, Match, and Emit pipeline that used vector embedding similarity, a deterministic penalty stack, and a multi-tier fallback cascade to generate cross-version page mappings that preserved navigation across releases.
- **Documentation Engineering & Strategy:** Automated headless browser screenshot capture to eliminate manual image maintenance, redesigned site information architecture around user journeys, and coauthored team processes, templates, and roadmaps.

Technical Writer · Entando · Remote — Oct 2021 – Mar 2023

Owned developer documentation for a Kubernetes-based application composition platform. Verified all content in cloud and Multipass VM environments through hands-on testing of micro frontend and microservice generation, NGINX and Strapi integrations, product UIs, and CLI tooling.

- Maintained a VuePress static documentation site with Markdown and Vue styling, and created the company glossary and style guide to standardize terminology across customer-facing materials.
- Authored an architectural white paper on Domain-Driven Design and Packaged Business Capabilities used to secure new investor funding, edited marketing and partner communications, and coached engineering contributors in documentation practices.

Hardware Validation Engineer · Amazon Lab126 · Sunnyvale, CA — Apr 2017 – Sep 2021

Led high-speed interface compliance testing (USB Type-C/PD, SATA, HDMI, eDP) from proof-of-concept through production certification for consumer hardware devices. Owned acceptance criteria, test plans, data collection, design feedback, and multi-vendor test environment bring-up across concurrent product programs.

- Architected and validated test stations for emerging interface standards, negotiating equipment purchases with Keysight, Rohde & Schwarz, and other vendors, securing budget approvals, and verifying station functionality ahead of silicon availability.
- Managed a joint initiative with Keysight and BitifEye to overhaul HDMI ARC/eARC test environments and compliance packages, providing debug data to resolve hardware and software incompatibilities and driving vendor configuration document revisions.
- Established USB Type-C and Power Delivery compliance test environments, identified false failures in vendor test sequences through systematic cross-vendor correlation, and engaged directly with the USB-IF Compliance Administrator and Chair of the BC1.2 Working Group to clarify specification requirements.
- Developed Python and test automation scripts for data collection and analysis, authored a 51-page internal Jitter Tolerance Validation Procedure reference, and provided cross-functional coverage for DDR compliance testing.

Senior Hardware Engineer · Oracle · Santa Clara, CA — Jul 2014 – Mar 2017

Led transmitter and receiver high-speed interface characterization for enterprise server silicon with an emphasis on jitter tolerance, signal quality, and PLL tuning. Optimized benchtop setups across BERTs, real-time and sampling oscilloscopes, TDR, and vector network analyzers (VNAs), including calibration, S-parameter measurement, and post-processing.

- Executed TX and RX characterization across duty cycle distortion (DCD), TX jitter, FIR equalization, voltage swing, CTLE, VGA, DFE, jitter tolerance (JTOL), and eye diagram signal quality, applying jitter tolerance methodologies to signal integrity analysis and IC design feedback.
- Developed automation scripts to control test equipment, sweep DUT parameters, and format measurement results, and performed schematic reviews, test board assembly, and component-level debug.

Senior Hardware Engineer · LSI Logic (now Broadcom) · Milpitas, CA — Apr 2008 – Jun 2014

Executed high-speed serial interface compliance testing and characterization across PCIe, SATA, SAS, Fibre Channel, SFP+, XFI, and IEEE 802.3 standards at data rates up to 32 Gbps. Served as an early adopter of 32 Gbps NRZ and PAM4 benchtop test solutions requiring complex multi-instrument calibrations and delay compensation.

- Contributed empirical data collection and test methodology experimentation to the IEEE 802.3bj standards committee, collaborating with standards bodies on compliance specification iterations.
- Developed custom instrument drivers and test algorithms to automate benchtop measurements, control DUT parameters, verify simulation models, and perform statistical analysis across validation programs.

Preceded by earlier roles at LSI Logic as Product Applications Engineer and Engineering Bench Technician (2004 – 2007) and Statistical Modeling Consultant at LipidX Technologies (2007 – 2008).

TECHNICAL PUBLICATIONS & OPEN-SOURCE ENGINEERING

- **Signal Integrity: From Waves to SerDes (21-Chapter Series & 2 Ebook Editions, 2026):** Authored Beginner and Advanced editions tracing electromagnetic wave propagation, transmission lines, S-parameters, jitter decomposition, equalization (CTLE, DFE, FFE), SerDes link budgets, PAM4, CDR, and coherent optical transmission physics, deriving every governing equation step by step and enforcing writing standards with a custom Python linter.
- **Transformer & Mixture of Experts Architecture Series (58 Chapters, 5 Ebooks, 2026):** Authored five interconnected series (Tokenization, Vector Embeddings, Positional Encoding & RoPE, Decoder-Only Transformers, and Sparse Mixture of Experts) plus a mathematical walkthrough of parallel training versus inference, deriving the forward pass, backpropagation, and top-k routing tensor algebra by hand.
- **AI Tooling Ecosystem & Reference Architecture Guides (Open-Source Templates & 19 Articles, 2026):** Authored a 7-part ecosystem mapping series and 6 two-part reference architecture and setup guides spanning LangGraph, LlamaIndex, MCP, LiteLLM, vLLM, SGLang, Weaviate, Pinecone, Qdrant, Langfuse, and Braintrust, plus an open-source cross-platform repository for coding agent governance across seven IDEs.
- **SERP API Benchmark for AI Overviews (Open-Source Python & Playwright Suite, 2026):** Built and open-sourced an empirical benchmark evaluating how reliably four SERP APIs extract Google AI Overviews for RAG pipelines across 50 queries, rebuilding the scoring methodology around detection yield to handle upstream non-determinism and auditing 196 raw responses for silent parser failures.

EDUCATION & RECOGNITION

B.S. Physics, University of California, Irvine — 2002 – 2006

Graduated with Honors (3.7 GPA) · UC Regents Scholar · Sigma Pi Sigma (National Physics Honors Society) · Campuswide Honors Program · Senior Thesis: Optical Modeling of Sub-Antarctic Neutrino Interaction (ASAP)

Winner, Document360 Global Writers Awards · recognized for overall body of technical documentation work — 2025

TECHNICAL SKILLS

TEST & MEASUREMENT:	Oscilloscopes (real-time and sampling), bit error rate testers (BERTs), arbitrary waveform generators (AWGs), vector network analyzers (VNAs), TDR, protocol analyzers, CDR units, compliance test suites, cross-vendor correlation
SIGNAL INTEGRITY & STANDARDS:	Jitter tolerance (JTOL), TX/RX characterization, eye diagram analysis, S-parameters, PAM4 and NRZ signaling, CTLE/DFE/FIR equalization, PCIe, IEEE 802.3bj, USB Type-C/PD, HDMI ARC/eARC, SATA/SAS, eDP, DDR
AI & MACHINE LEARNING:	Transformer architecture (embeddings, RoPE, multi-head attention, backpropagation), Mixture of Experts (MoE), agentic pipelines, LLM orchestration, RAG, semantic search, fidelity scoring, model evaluation, MCP
LANGUAGES & AUTOMATION:	Python (NumPy, pandas, matplotlib), MATLAB, C, Playwright, pytest, Bash/Zsh, YAML, JSON, Markdown, Git/GitHub, instrument drivers, test automation scripting
DATA & STATISTICAL ANALYSIS:	Statistical modeling, parameter sweep optimization, regression analysis, instrument and output correlation, threshold calibration, benchmark design, k-anonymity privacy modeling
COMMUNICATION & INFRA:	Solution architecture presentations, technical ebooks, white papers, API and developer documentation, docs-as-code, SME interviews, Docker, Kubernetes, Linux CLI